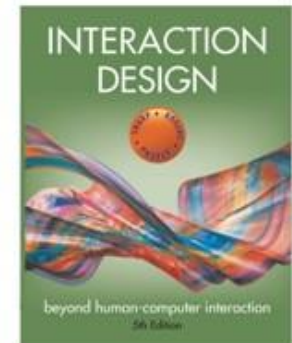


Topics covered last lecture (Lecture 4)

- **Interfaces**

- Metaphors, interface types, choosing interfaces to match user needs
- This part of the module is taken from the main text book for the module, the latest release from 2019. This lecture focuses on chapter 7.

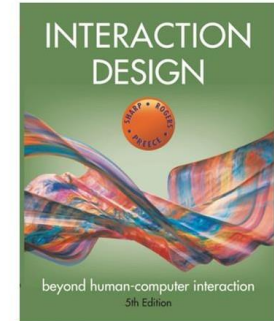


Chapter 7 INTERFACES

Topics covered this lecture (Lecture 5)

Prototyping

- Conceptual design, Concrete design, Using scenarios, Generating prototypes, Storyboard, User experience maps, Customer journeys
- This part of the module is taken from the main textbook for the module, the latest release from 2019. This lecture focuses on chapter 12.



Chapter 12

DESIGN, PROTOTYPING and
CONSTRUCTION

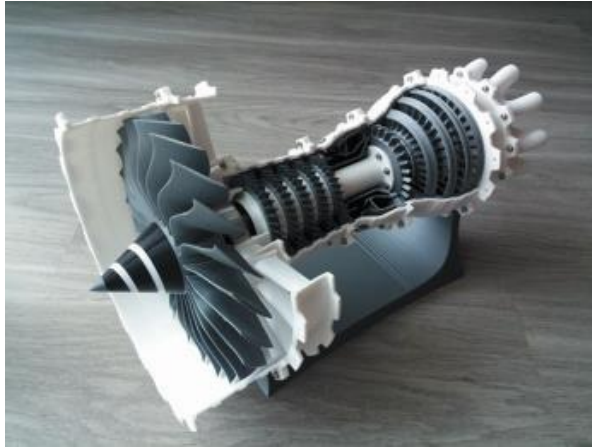
What is a prototype?

- One **manifestation** of a design that allows stakeholders to **interact** with it
- In other design fields, a **prototype** is a small-scale model:
 - A **miniature car**
 - A **miniature building or town**

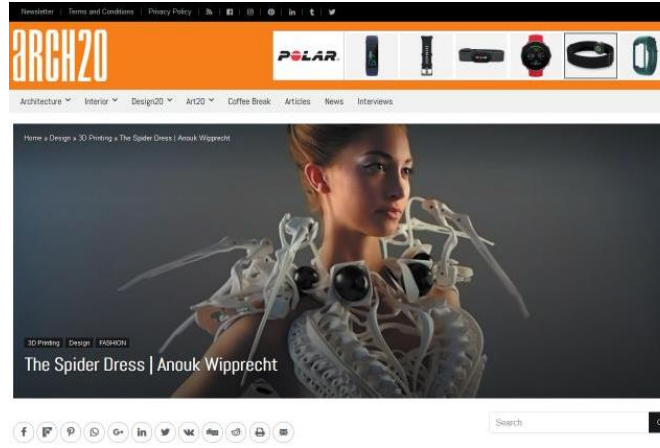
Source: [PalmPilot wooden model](#)
© Mark Richards



3D Printing Examples



(a)



(b)



(c)

Examples of 3D printing: (a) model jet engine, (b) Spider Dress 2.0 by Anouk Wipprecht, and (c) teddy bear “printed” from a wireframe design

Sources: (a) [Build Your Own Jet Engine](#). Licensed under CC-BY-3.0, (b) [arch20](#), and (c) used courtesy of Scott Hudson

What is a prototype in interaction design?

In interaction design, a prototype can be (among other things):

- A series of screen **sketches**
- A **storyboard**, for example, a cartoon-like series of scenes
- A **PowerPoint** slide show
- A **video** simulating the use of a system
- A **physical object** (for instance, a lump of wood in the PalmPilot)
- A **cardboard mock-up**
- A piece of **software** with limited functionality written in the target language or in another language

Why prototype?

- **Evaluation** and **feedback** are **central** to interaction design
- **Stakeholders** can see, hold, and interact with a prototype more easily than a document or a drawing
- Team members can **communicate** effectively
- **Ideas can be tested** out
- Prototyping **encourages reflection**: an important aspect of design
- Prototypes **answer questions** and support designers in choosing between alternatives

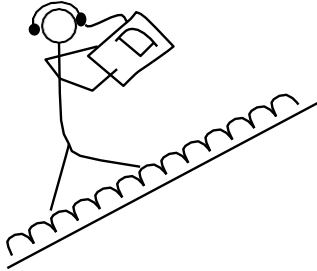
Low-fidelity Prototyping

- Uses a **medium** which is **unlike** the final medium, for example, paper or cardboard
- Is quick, cheap, and easily changed
- We discuss the following examples in more detail:
 - ‘Post-it’ notes (except this one as its quite obvious)
 - Storyboards
 - Sketches of screens, task sequences, and so on
 - ‘Wizard-of-Oz’

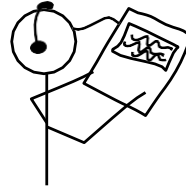
Storyboards

- It is a **series of sketches** showing how a user **might** progress through a **task** using the product
- Often **used with scenarios**, bringing in more detail and a chance to **role play**

Example storyboard



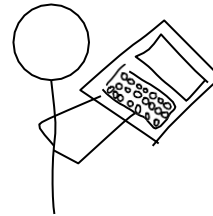
Christina walks up hill; the product gives her information about the site



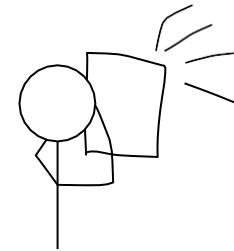
Christina adjusts the preferences to find information about the pottery trade in Ancient Greece



Christina scrambles to the highest point



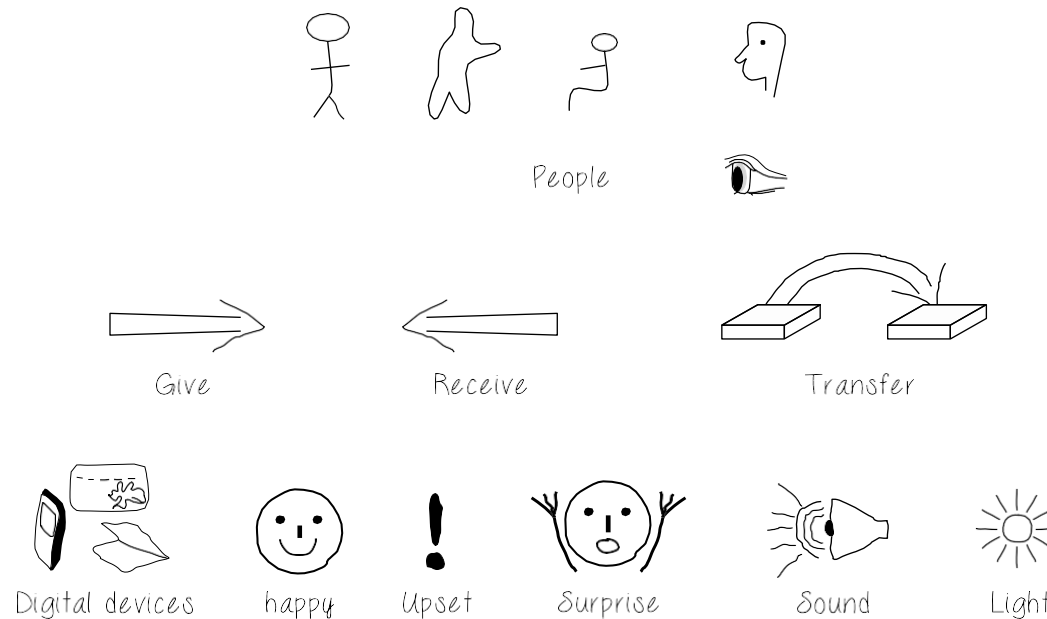
Christina stores information about the pottery trader's way of life in Ancient Greece



Christina takes a photograph of the location of the pottery market

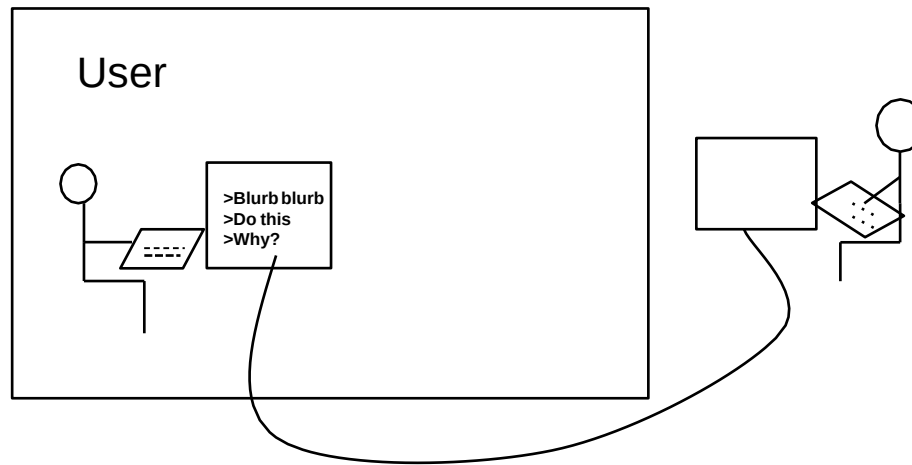
Sketching

- Low-fidelity prototyping often **relies on sketching**
- **Don't be inhibited about drawing ability**
 - Practice simple symbols
 - You don't have to be an artist!



‘Wizard-of-Oz’ prototyping

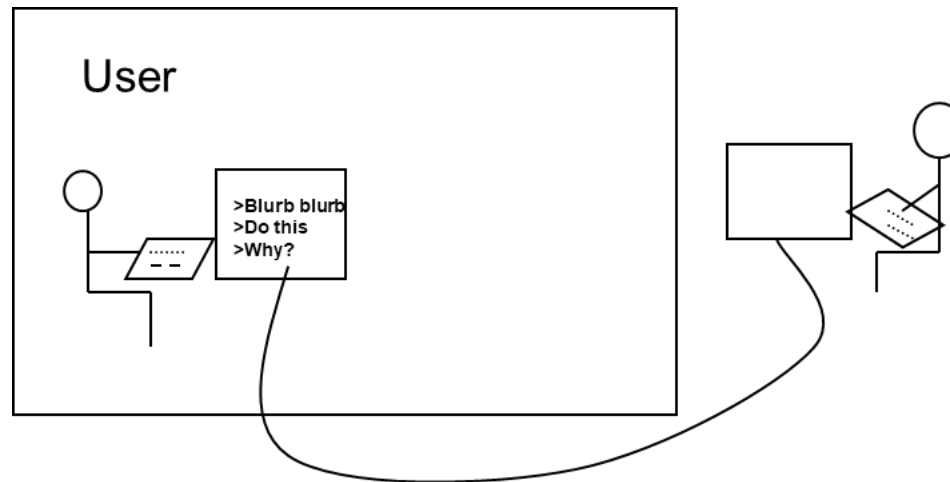
- The user **thinks** they are **interacting with a computer**, but a **human is responding** to output rather than the system
- Usually done early in design to understand users’ expectations
- Is there anything ‘wrong’ with this approach?



‘Wizard-of-Oz’ prototyping

This methodology requires three things:

- **a script** that provides directions for what is to take place,
- **a participant** to play the role of the end user and
- **a human “wizard”** that will perform tasks that will simulate the behaviour of the completed product.



Issues with deception

- The person playing the end user may/may not know that they are playing a role or that the wizard's tasks are being performed manually by a human being instead of by a machine or computer program.

Faculty of Science and Engineering: Guidelines for Research Ethical Approval (November 2020)

Use of deception

Research studies requiring deception of participants will not normally be approved. Applicants wishing to use deception are advised to submit detailed arguments. Use of deception inevitably limits informed consent which must then be obtained retrospectively at the debriefing stage.

Where any kind of deception has been used, full and frank debriefing is essential. Any subject who objects to the procedure has the right to have their data destroyed and withdrawn. This right is an inevitable consequence of the lack of prior informed consent which results from deception. Deception at the debriefing stage is absolutely unacceptable.

High-fidelity prototyping

- Uses **materials** that you **would expect** to be in the final product
- Prototype looks **more like the final system** than a low-fidelity version
- High-fidelity prototypes can be developed by integrating existing hardware and software components
- Danger that users think they have a complete system...see compromises

Compromises in prototyping

- Prototyping involve **compromises**
 - For software-based prototyping, maybe there is a slow response? sketchy icons? limited functionality?
 - “In the wild” prototypes operational but not necessarily robust
- Two common types of compromise:
 - **Horizontal**: Provides a wide range of functions, but with little detail
 - **Vertical**: Provides a lot of detail for only a few functions
- **Compromises in prototypes must not be ignored.**

Conceptual design

- A **conceptual model** is an outline of what people can **do** with a product and what **concepts** are needed to understand and interact with it
- Understand **problem space** and current requirements
 - empathize with users
 - This is **often rushed** and may lead to issues later on
- Creativity and brainstorming techniques
- **Mood board** may capture desired feel

<https://itcraftapps.com/blog/moodboard-mobile-app-design-inspiration/>

- Consider alternatives: **scenarios** and **prototyping** helps

PERSONA 1 - SCENARIO 1



SCENARIOS

UX Knowledge Base Sketch #16

PERSONA 2 - SCENARIO 1



SCENARIOS ARE

-  SHORT STORIES ⇒  SEQUENCE OF ACTIONS AND EVENTS FROM THE MAIN CHARACTER'S (PERSONA) PERSPECTIVE.
-  ABOUT A PERSONA ⇒  BASED ON THE PERSONA'S NEEDS, GOALS, MOTIVATIONS
-  TRYING TO ACHIEVE HER GOAL ⇒  BEFORE IMPLEMENTING SG, ASK: WOULD THIS BE USEFUL TO OUR USERS?
-  BY USING THE FUTURE PRODUCT / SERVICE ⇒  IT IS ABOUT THE FUTURE!
-  IN HER CONTEXT. ⇒  REAL WORLD CONTEXT CONTAINING INFORMATION ABOUT BEHAVIOR, USAGE, CONDITIONS AFFECTING THE EXPERIENCE.

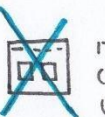
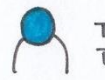
SCENARIO $\neq$ USE CASE

-  LIKE A TECHNICAL BLUEPRINT
-  IT IS ABOUT FUNCTIONS, NOT THINKING OR FEELINGS
-  LIST OF ACTIONS BETWEEN THE USER AND THE SYSTEM

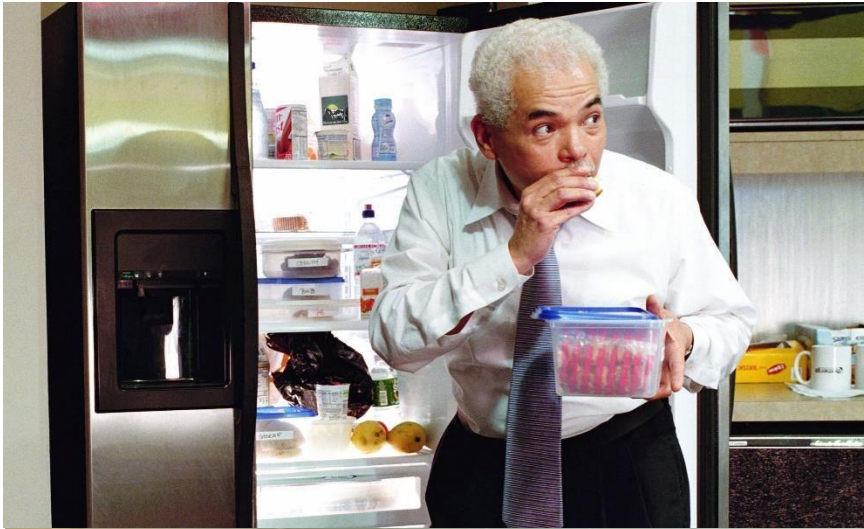
|| SCENARIO $\neq$ USER STORY

-  SHORTER (SOME SENTENCES)
-  DESCRIBING A NEED FROM A PERSONA'S PERSPECTIVE
-  IN THE PERSONA'S OWN WORDS.

HOW TO CREATE SCENARIOS

- ☒ PREREQUISITE: PERSONAS
-  DECIDE WHAT SCENARIOS SHOULD BE CREATED: MOST SIGNIFICANT ACTIVITIES BY EACH PERSONA
-  MAKE A BELIEVABLE STORY
 - ↳ PLOT: TRIGGER, ACTIONS, ACHIEVED GOAL
 - ↳ CONTEXT, E.G.: WHERE?
WEATHER CONDITIONS
LIGHT CONDITIONS
WHEN?
 - ↳ MOTIVATIONS, REASONS
 - ↳ HOW THE FUTURE PRODUCT / SERVICE CAN HELP REACHING THE END-GOAL?
-  IT IS ABOUT THE PERSONA'S EXPERIENCE (THINKING, FEELING, SEEING, DOING), NOT ABOUT UI SOLUTIONS!
-  TRY TO AVOID UNREALISTIC ASSUMPTIONS! THINK ABOUT WHAT THE PERSONA WOULD DO!
-  EXCLUDE THE UNNECESSARY SENTENCES!
- ⇒ THEN YOU CAN START DEFINING REQUIREMENTS FROM THE SCENARIOS!

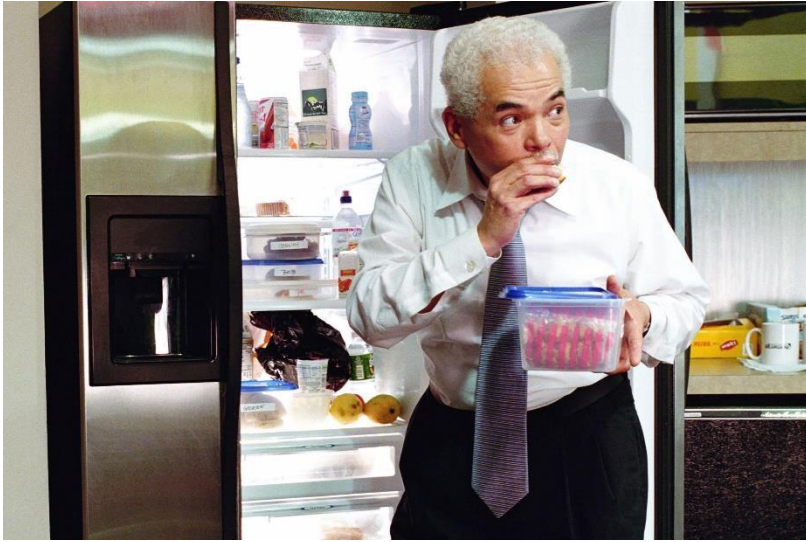
Building scenarios



You can **almost** 'feel' the **tension** here ...



Building scenarios



- This has a **different 'feel'**
- Makes you smile
- A **light atmosphere** of 'being caught' here?
- You can **use images to set your scene/mood** etc.
- Scenarios are **built upon** user stories but scenarios are not equal to user stories.
- They **take User Stories to the next level** by adding the interaction with the product or service to the story.

Concrete design

- Difference between **conceptual** and **concrete** is emphasis
- **Many aspects** to concrete design
 - Color, icons, buttons, interaction devices, and so on
- User **characteristics** and **context**
 - Inclusiveness, input, and output modes
- **Accessibility**
 - Web Content Accessibility Guidelines
- **Cross-cultural design**
 - Language, colors, icons, and information architecture
 - Date format? dd/mm/yyyy in UK vs mm/dd/yyyy in USA
 - Indigenous knowledge and perspectives

Generating prototypes

- Generate a **storyboard** from a scenario
 - **Break down** scenario into **steps**
 - Create a **scene** for each **step**
- Sketching out a storyboard **prompts** designers to think about **design issues**
- **You can** use software to do this

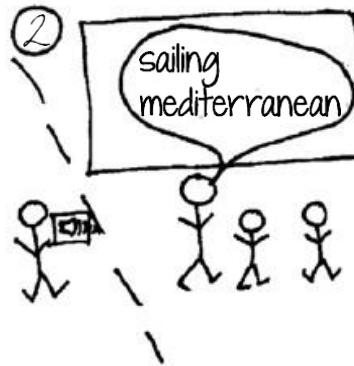
<https://blog.hubspot.com/marketing/storyboard-software>

- **You will** be asked to generate a **card-based prototype** from a storyboard or from a use case
 - **Consider** each step in use case – what interaction element is needed
 - **Draw** a card that captures it

Generating a storyboard



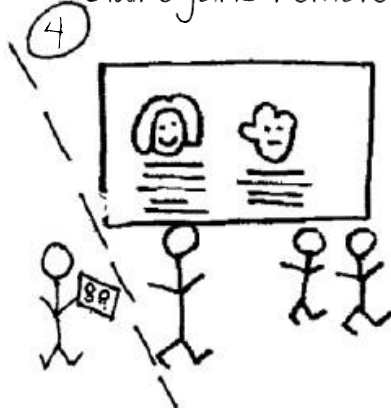
Thomson family
gather around
Claire joins remotely



Will tells their
initial idea



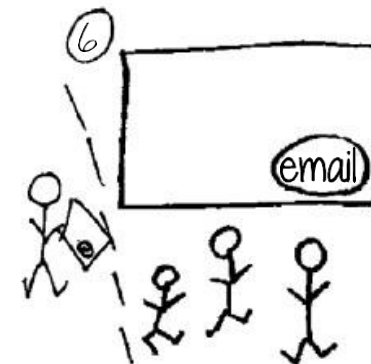
System suggests
flotilla



System shows
descriptions



Will asks
for details



Details emailed

Generating a card-based prototype



<http://aaronbrako.com/prototyping>

Paper prototypes

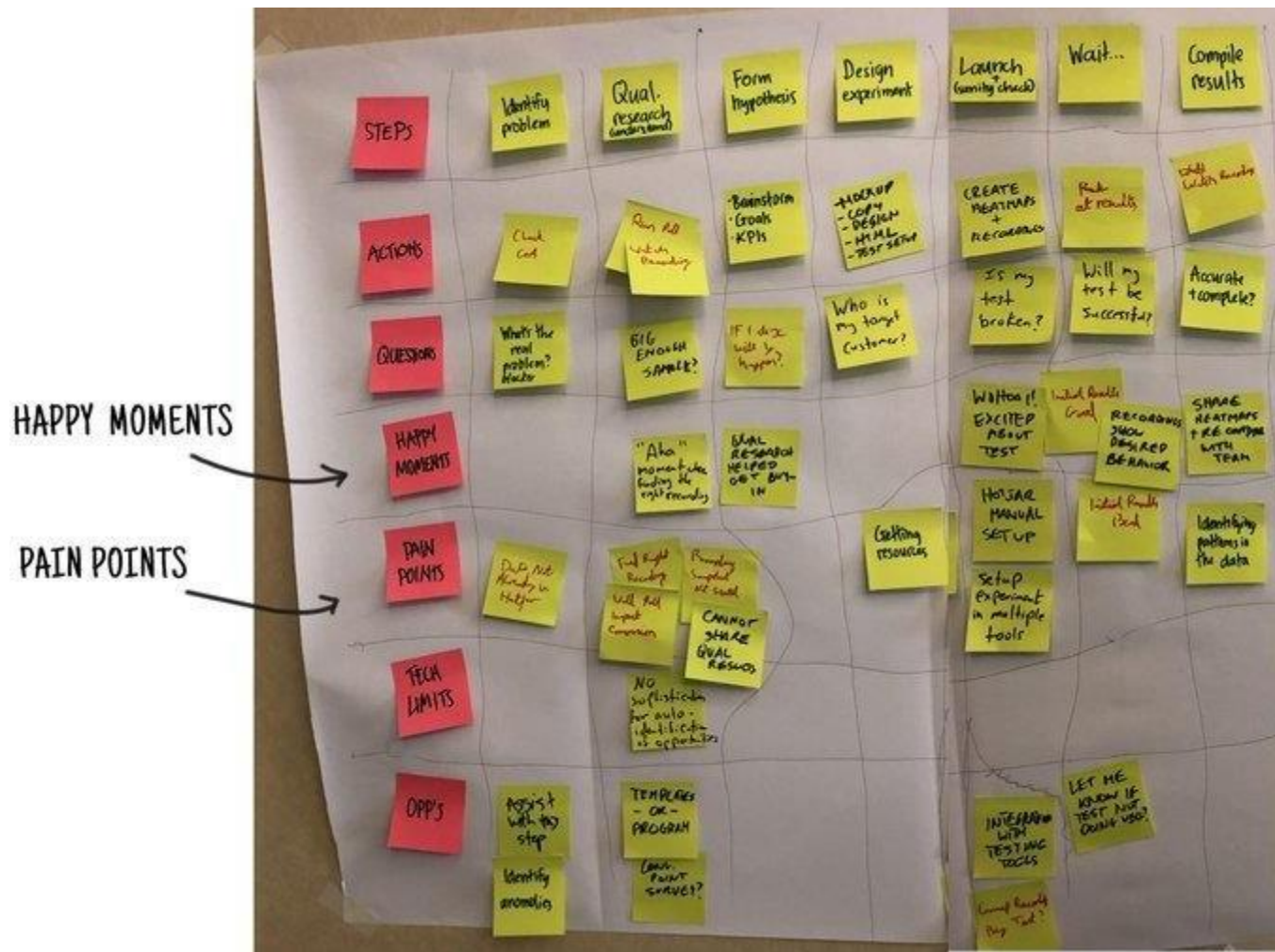


<https://www.youtube.com/watch?v=yafaGNFu8Eg>

Explore the user's experience

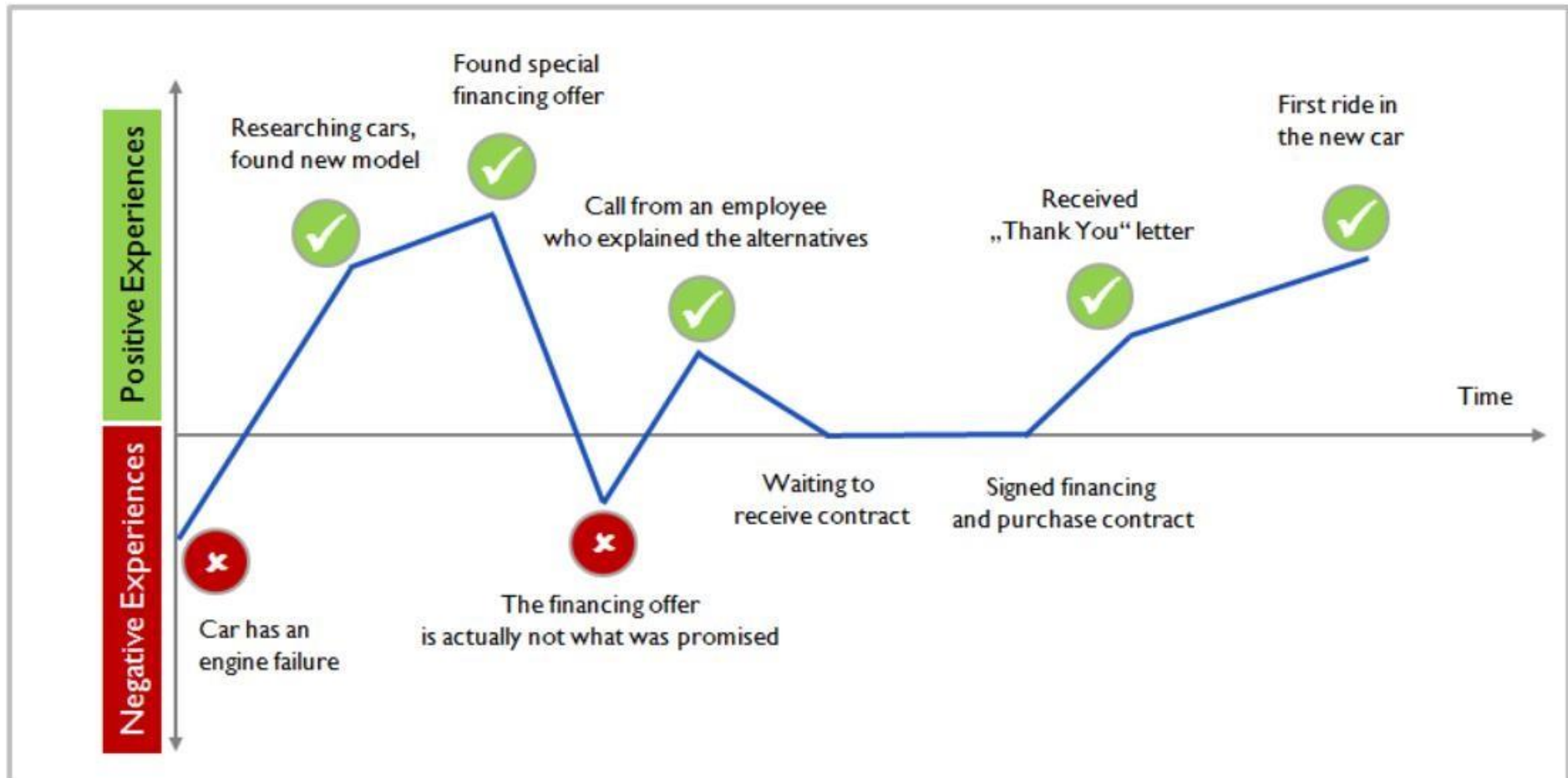
- Use card-based **prototypes** to model the user experience
- Visual representation called:
 - Customer or **user journey map**
 - Coming up ... car purchase example uses timeline
 - **Experience map**
 - Coming up ... the wheel of experience
- Two common representations
 - **Wheel**
 - **Timeline**

Customer journey map using post-its



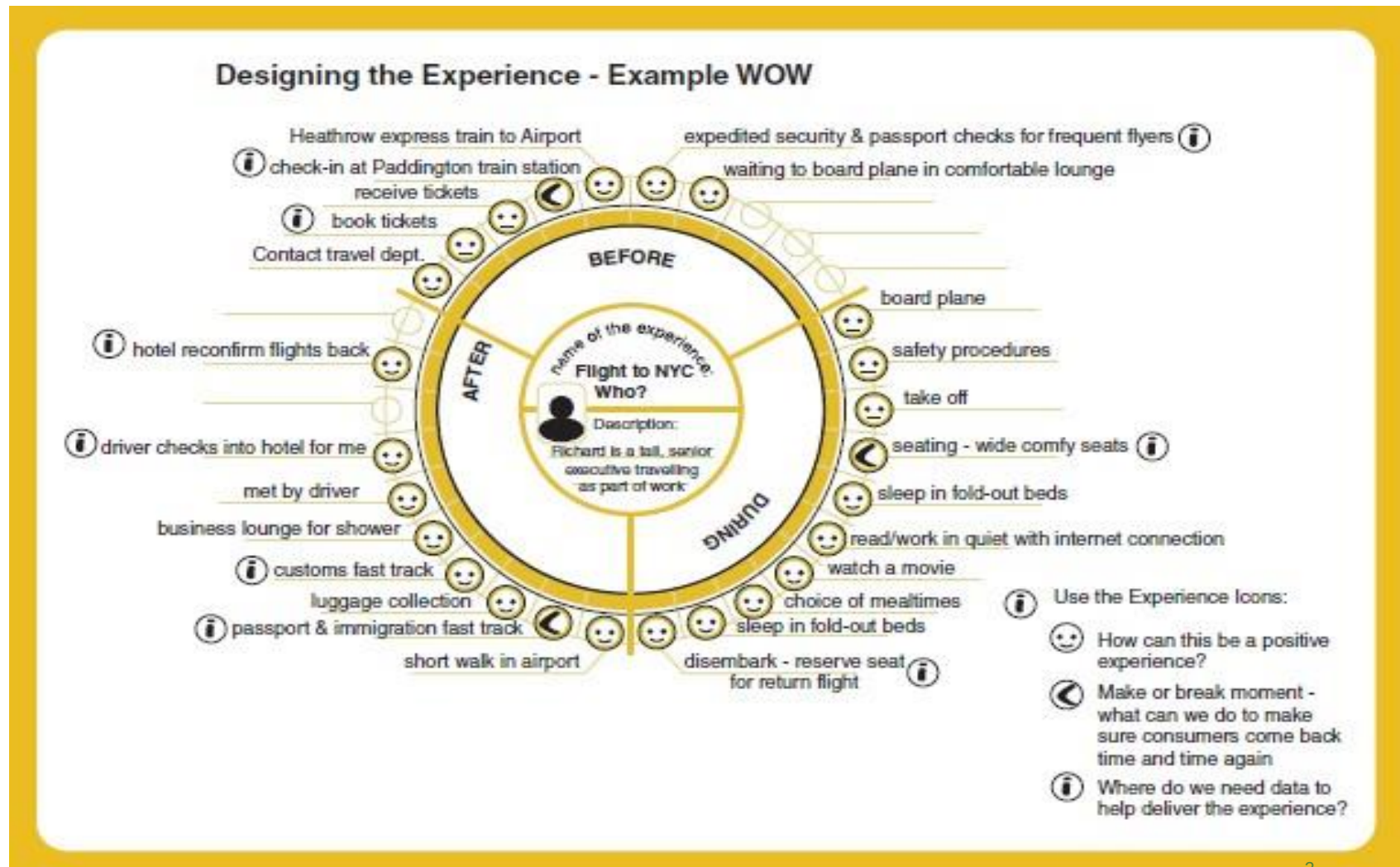
<https://www.hotjar.com/customer-journey-map/>

Customer Journey Example: Car Purchase

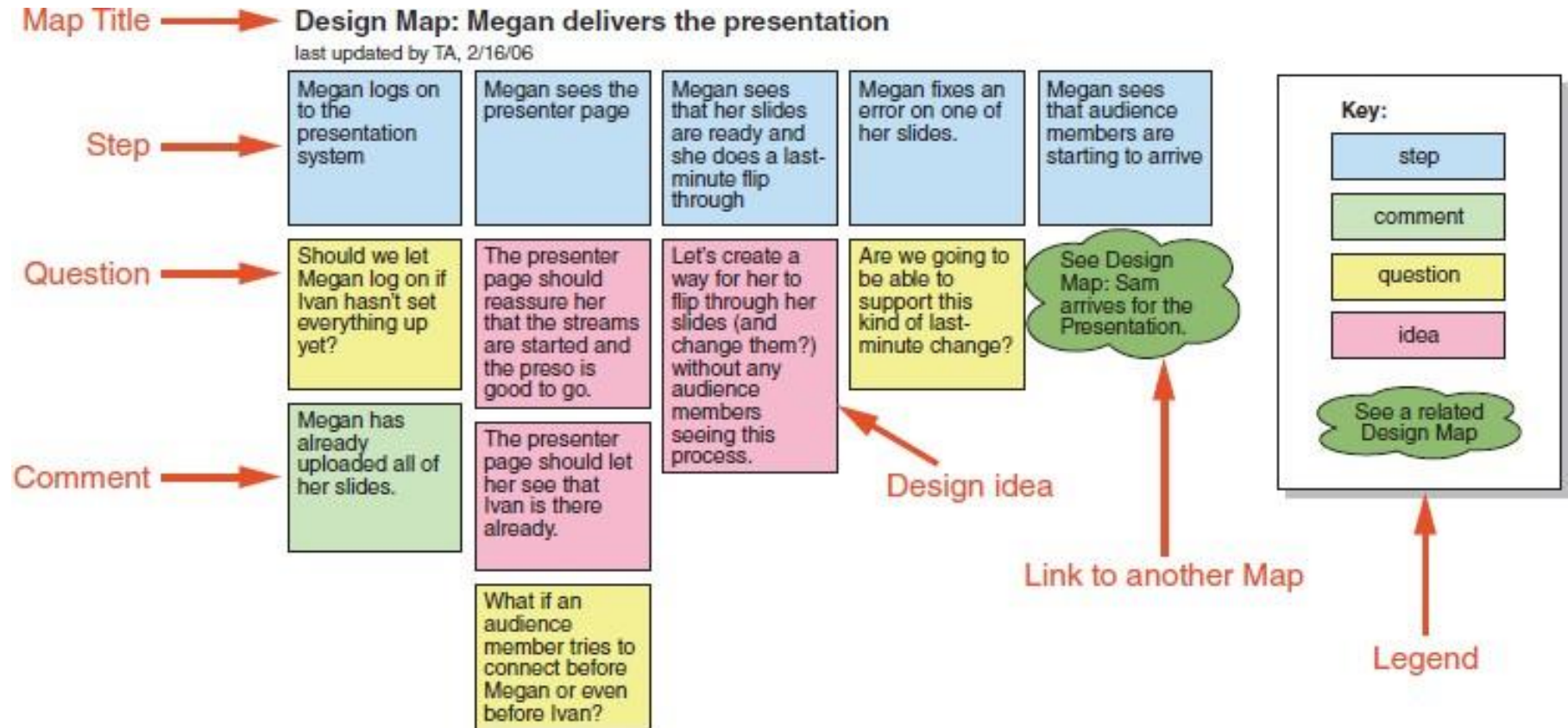


<http://usability-ed.blogspot.com/2011/11/mapping-customer-journeys.html>

User experience map drawn as a wheel



An experience map drawn as a timeline

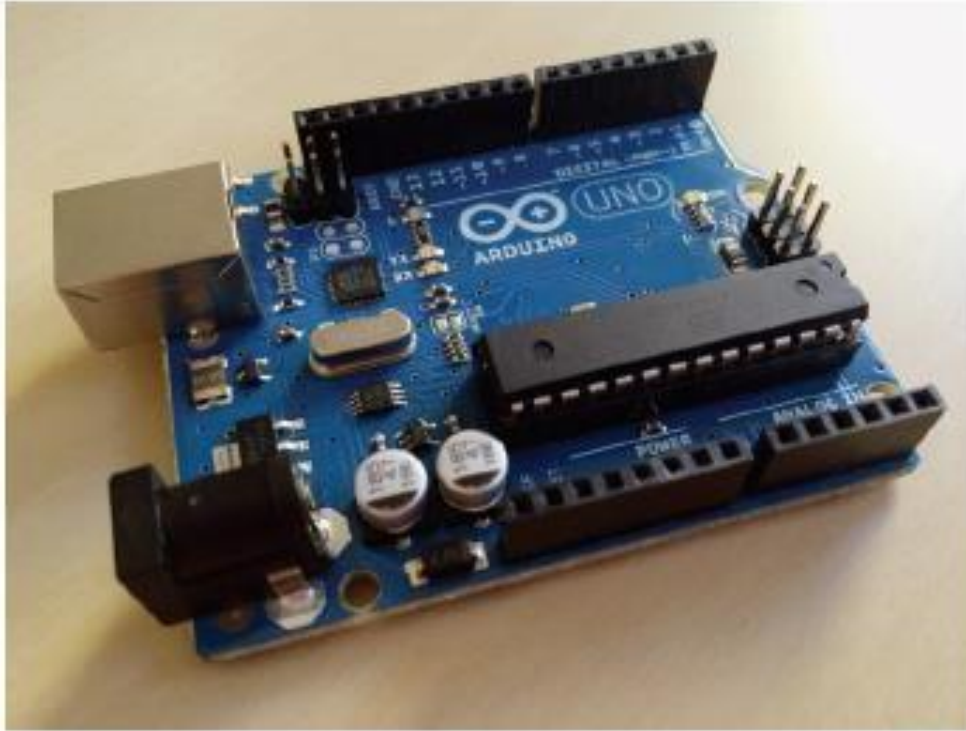


Source: Adlin and Pruitt (2010), p134. Used courtesy of [Morgan Kaufmann](#).

Construction: Physical computing

- Build and code prototypes using electronics
- Toolkits available include
 - Arduino
 - LilyPad (for fabrics)
 - Senseboard
 - BBC micro:bit
 - MaKey MaKey
- Designed for use by wide range of people

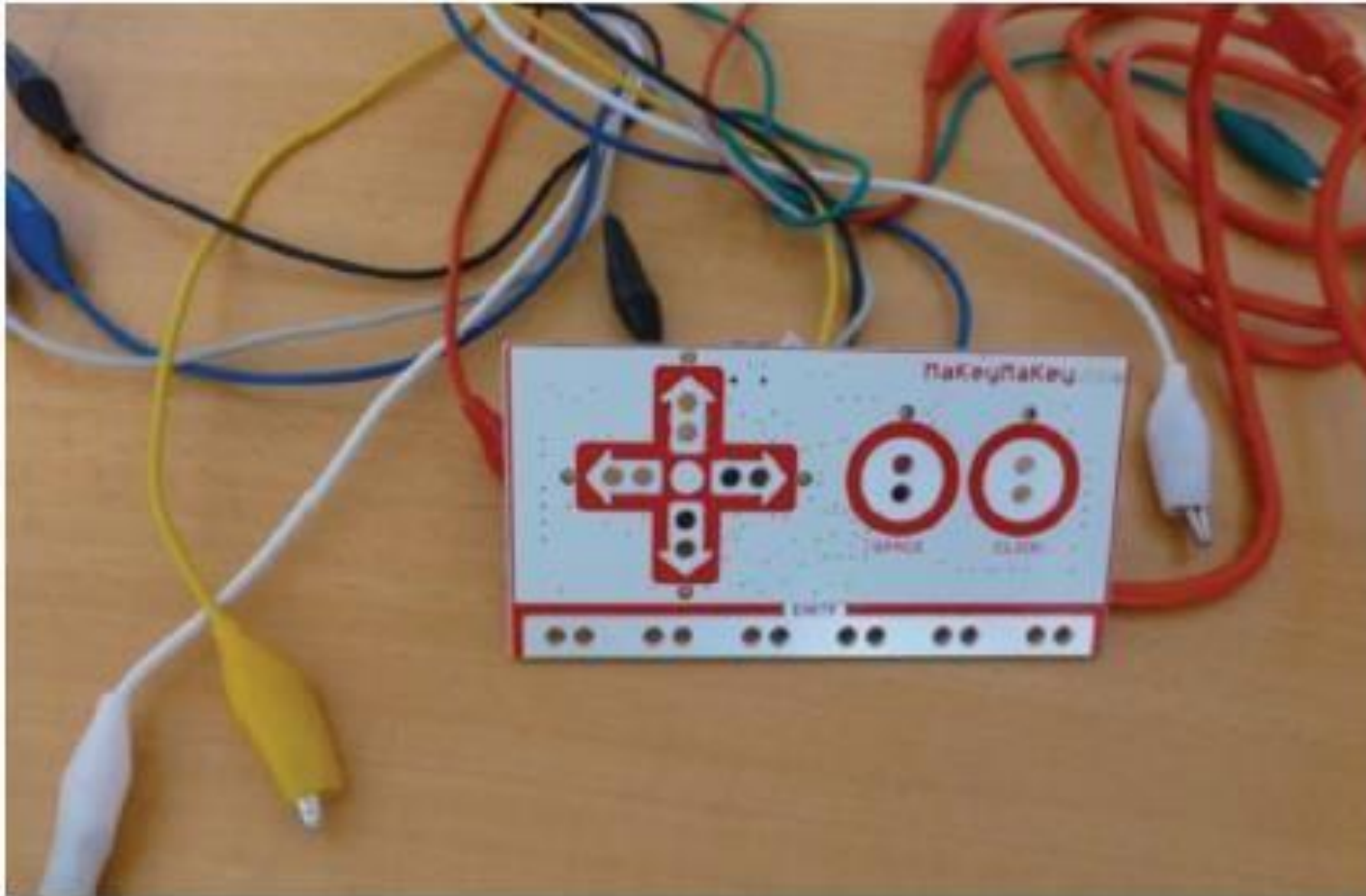
Physical computing kits



The Arduino board

Source: Used courtesy of Dr. Nicolai Marquardt

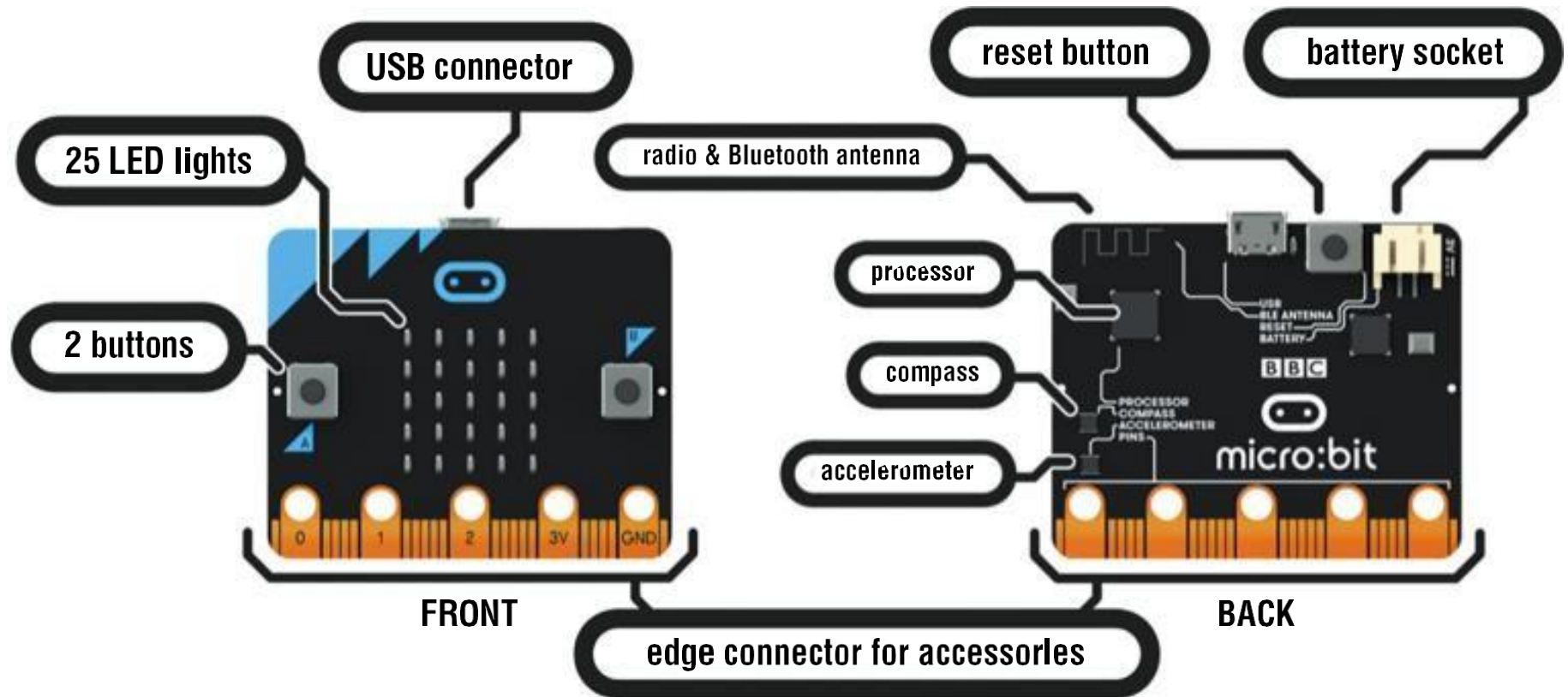
Physical computing kits



The Makey Makey toolkit

Source: [Makey Makey](https://makeymakey.com/)

Physical computing kits



The BBC micro:bit

Source: [micro:bit](https://microbit.org/). Used courtesy of Micro:bit Foundation

Construction: SDKs

- **Software Development Kits**
 - Programming tools and components to develop for a specific platform, for example, iOS
- Includes: **IDE, documentation, drivers, sample code, and application programming interfaces (APIs)**
- Makes development much easier
- Examples:
 - **Amazon's Alexa Skills Kit** for voice-based services
 - Apple's **ARKit** for augmented reality
 - **Microsoft's Kinect SDK** for motion tracking

Thank you